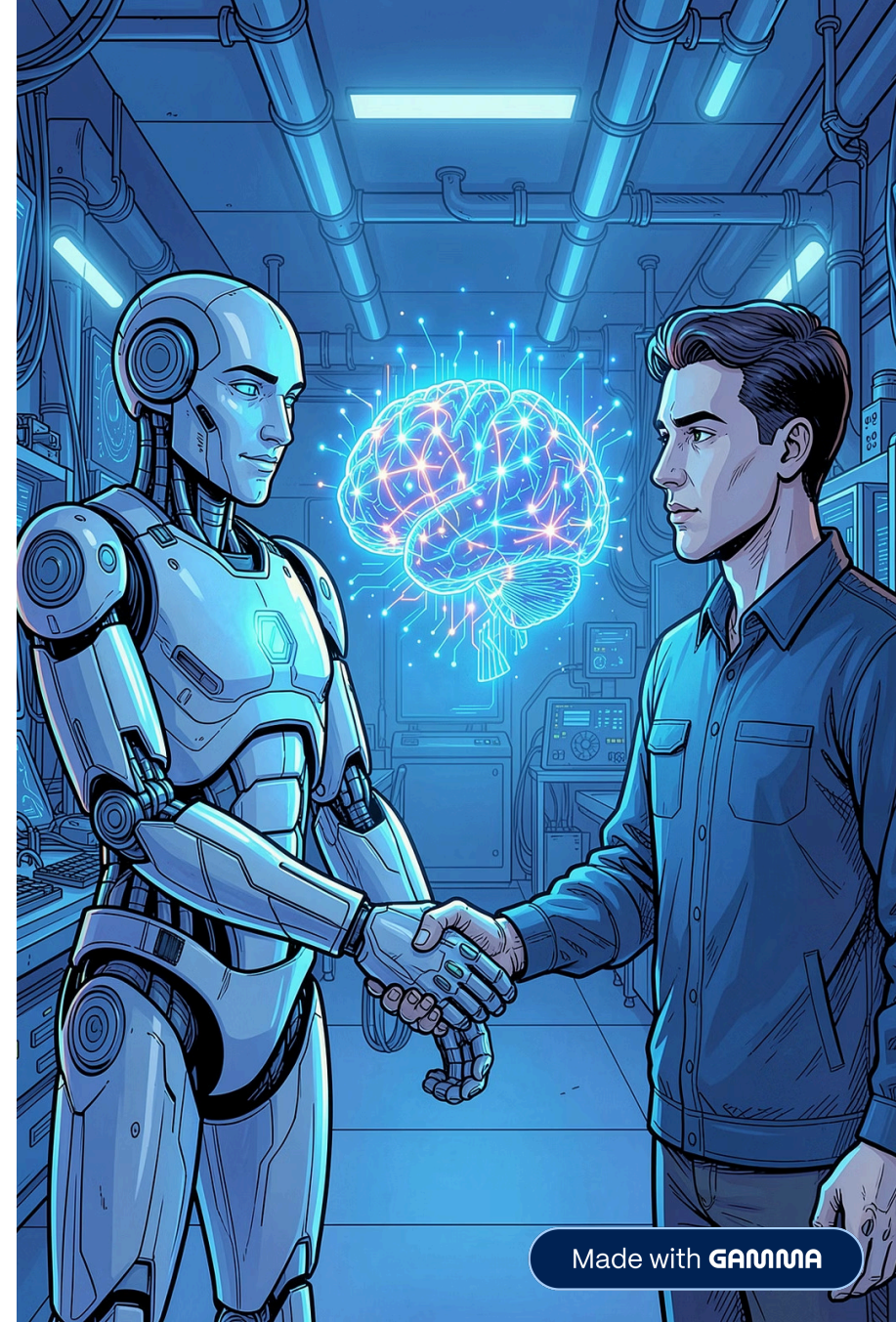


UNIT 1 · FOUNDATIONS OF AGENTIC AI

Intelligent Agents & Agent-Environment Interaction

A 30-minute introduction to how autonomous AI systems perceive, reason, and act — the foundation of every modern agentic system.

Presented by **Harshal Raut**



Made with **GAMMA**

Can ChatGPT Book Your Flight?



Try This Thought Experiment

Almost everyone has used ChatGPT. But can it **independently** book your flight, compare hotel prices, pay online, and send you an itinerary – without step-by-step instructions?

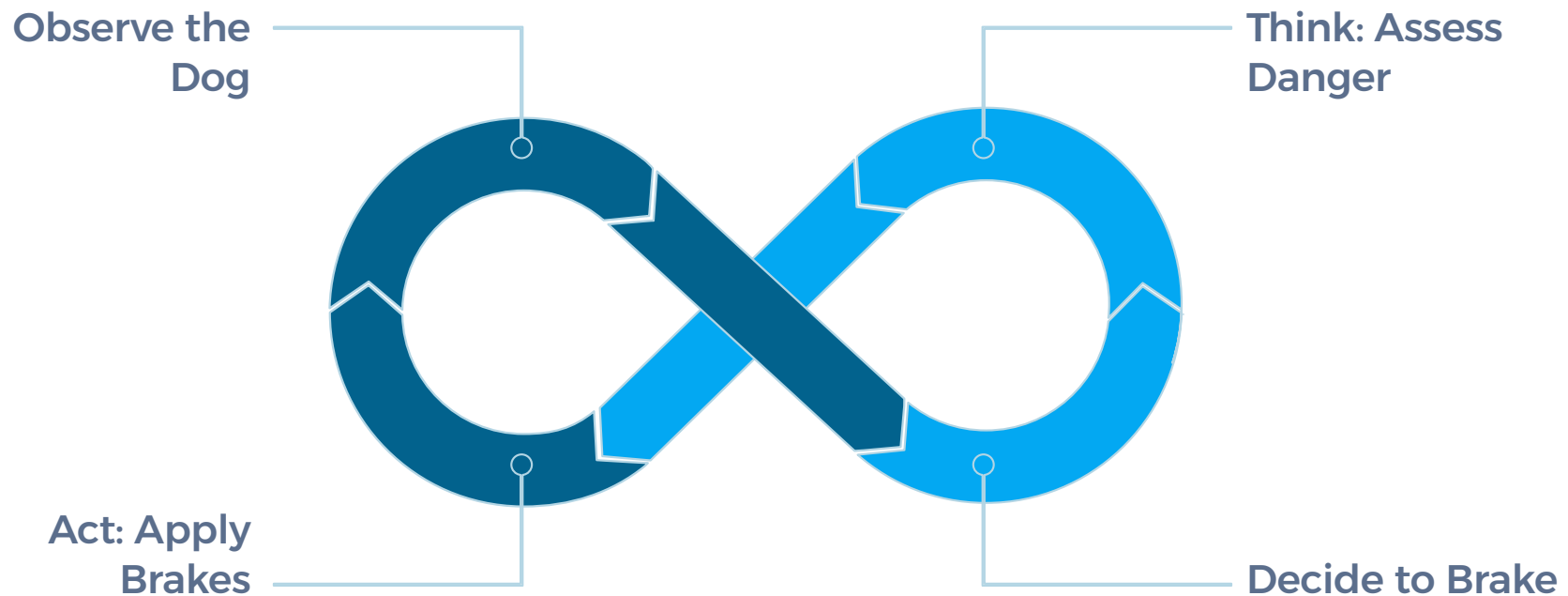
No. ChatGPT generates text. Acting autonomously requires something more.

- ❑ That "something more" is what makes an **Intelligent Agent**. That's today's topic.

CORE CONCEPT

What Is Intelligence?

Imagine riding your bike when a dog suddenly crosses your path. You don't brake immediately — your brain runs a loop first.



This cycle — **observe** → **think** → **decide** → **act** — is what intelligence looks like in action. Tesla's cameras observe the road, its AI decides, and its brakes respond. **Can machines do the same?**

DEFINITION

What Is an Intelligent Agent?

An **Intelligent Agent** is an autonomous system that **perceives** its environment, **reasons** about it, **decides**, and **acts** to achieve specific goals.



Autonomous

Acts without constant human input



Goal-Driven

Works toward defined objectives



Rational

Chooses the best available action



Reactive

Responds to real-time changes



Adaptive

Improves from experience over time

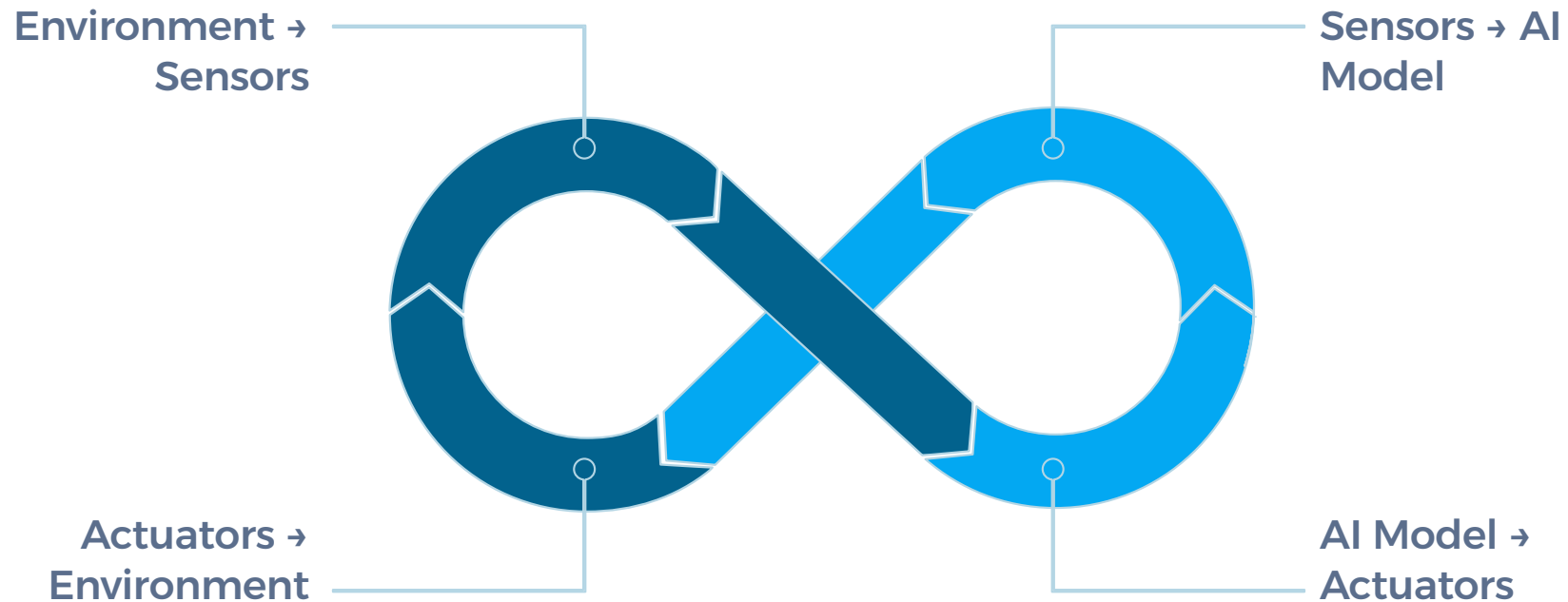


Learning

Gets better with feedback

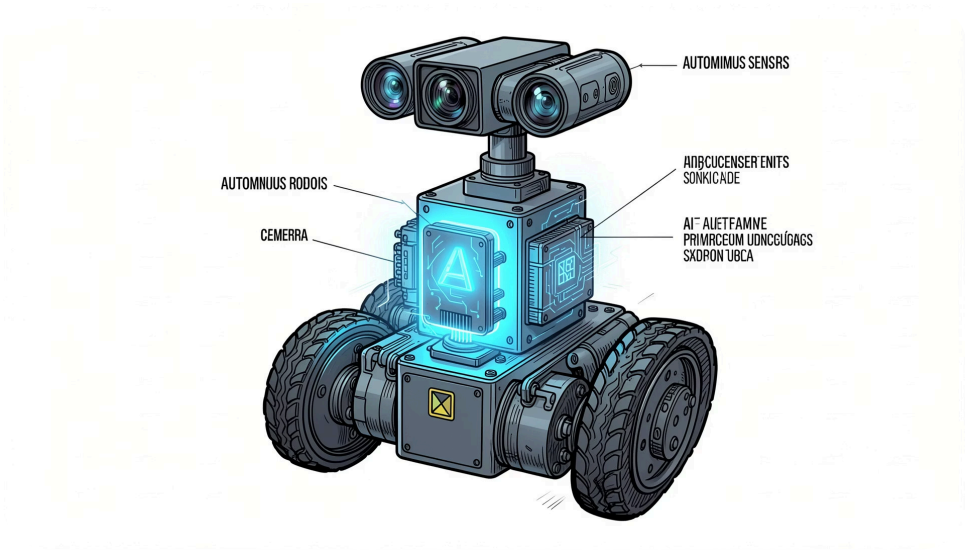
CORE MODEL

The Agent-Environment Interaction Loop



This loop **never stops**. The environment changes → sensors detect it → AI reasons → actuators respond → the environment changes again. This continuous cycle is the heartbeat of every autonomous system – from a robot vacuum to a self-driving car.

Components of an Intelligent Agent

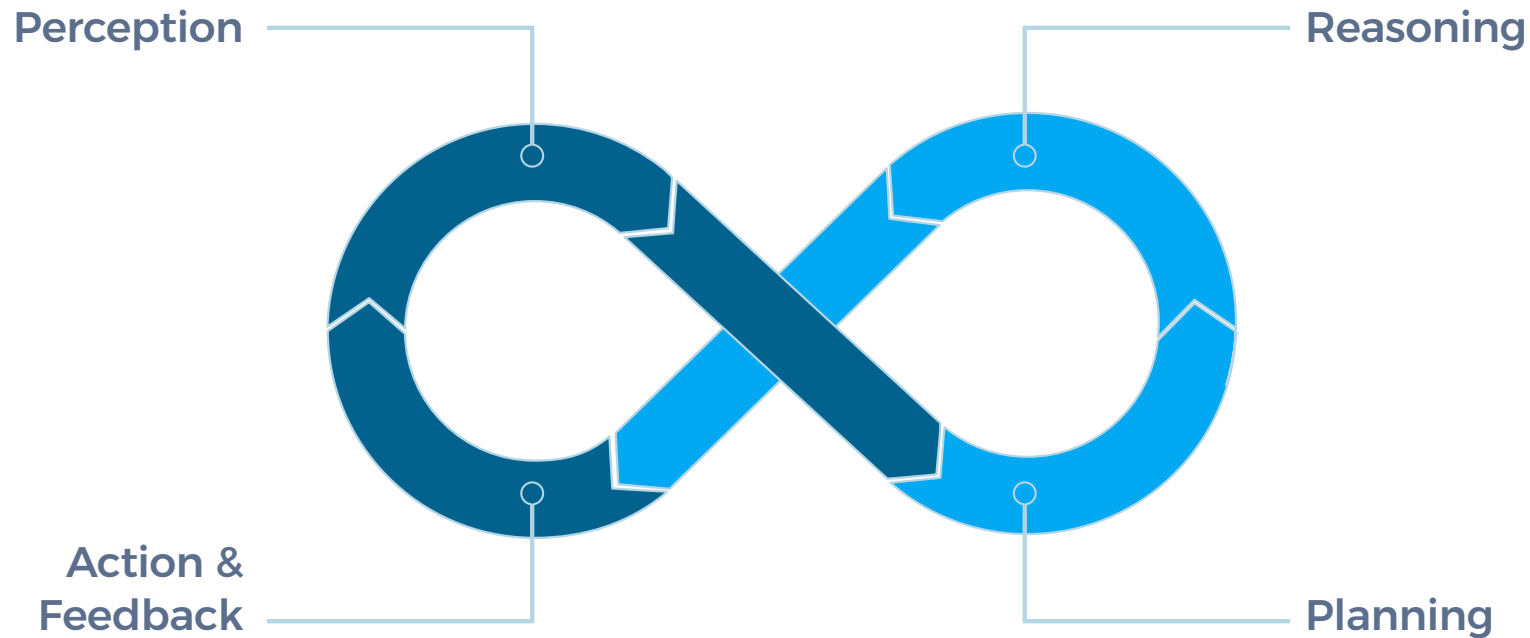


Component	Role	Example
Environmen t	Everything surrounding the agent	Road, home, hospital
Sensor	Collects raw data	Camera, GPS, radar
AI Model	Processes data, makes decisions	Neural network
Actuator	Executes physical or digital action	Motor, brake, speaker

Without sensors, the agent is **blind**.
Without actuators, it is **powerless**.

THE CORE LOOP

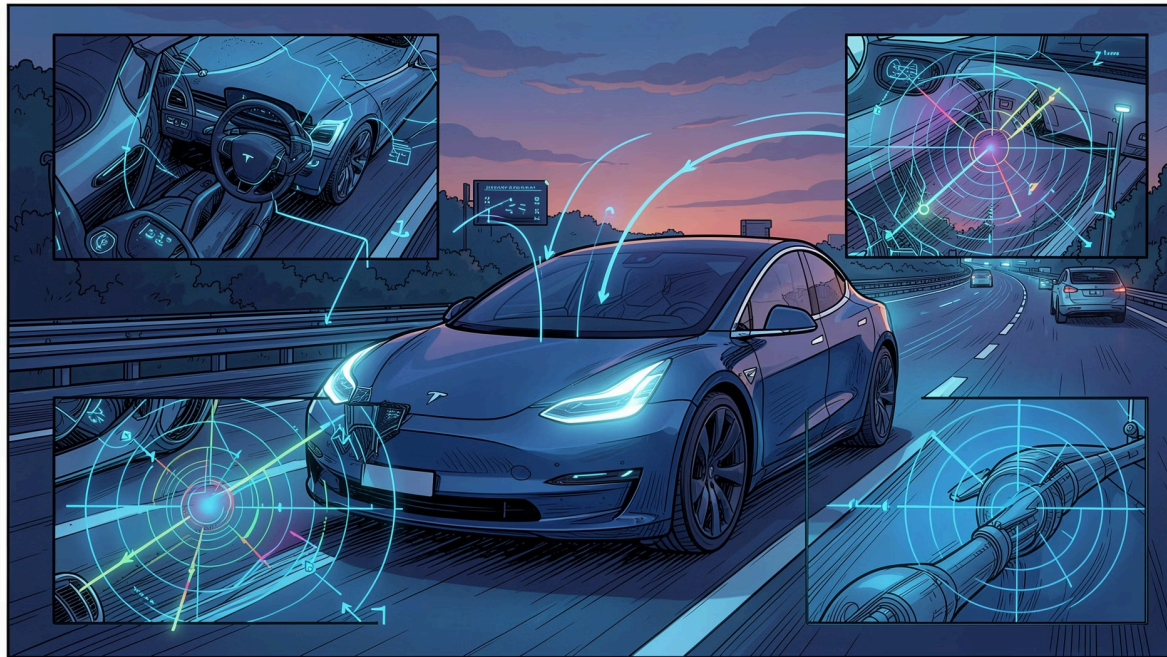
Perception → Reasoning → Action



This is the **heart of Agentic AI**. A food delivery system perceives an order, reasons about restaurant options, plans a route, assigns a driver, and uses customer ratings as feedback to improve. **The loop never stops – it continuously refines itself.**

CASE STUDY

Tesla: Intelligence in Motion



How Tesla's Agent Works Every Second

→ Perceive

Cameras, radar, and GPS stream thousands of data points per second

→ Reason

AI model analyzes the scene – detects obstacles, lane markings, speed

→ Act

Brakes, steering, and throttle respond in real time

→ Repeat

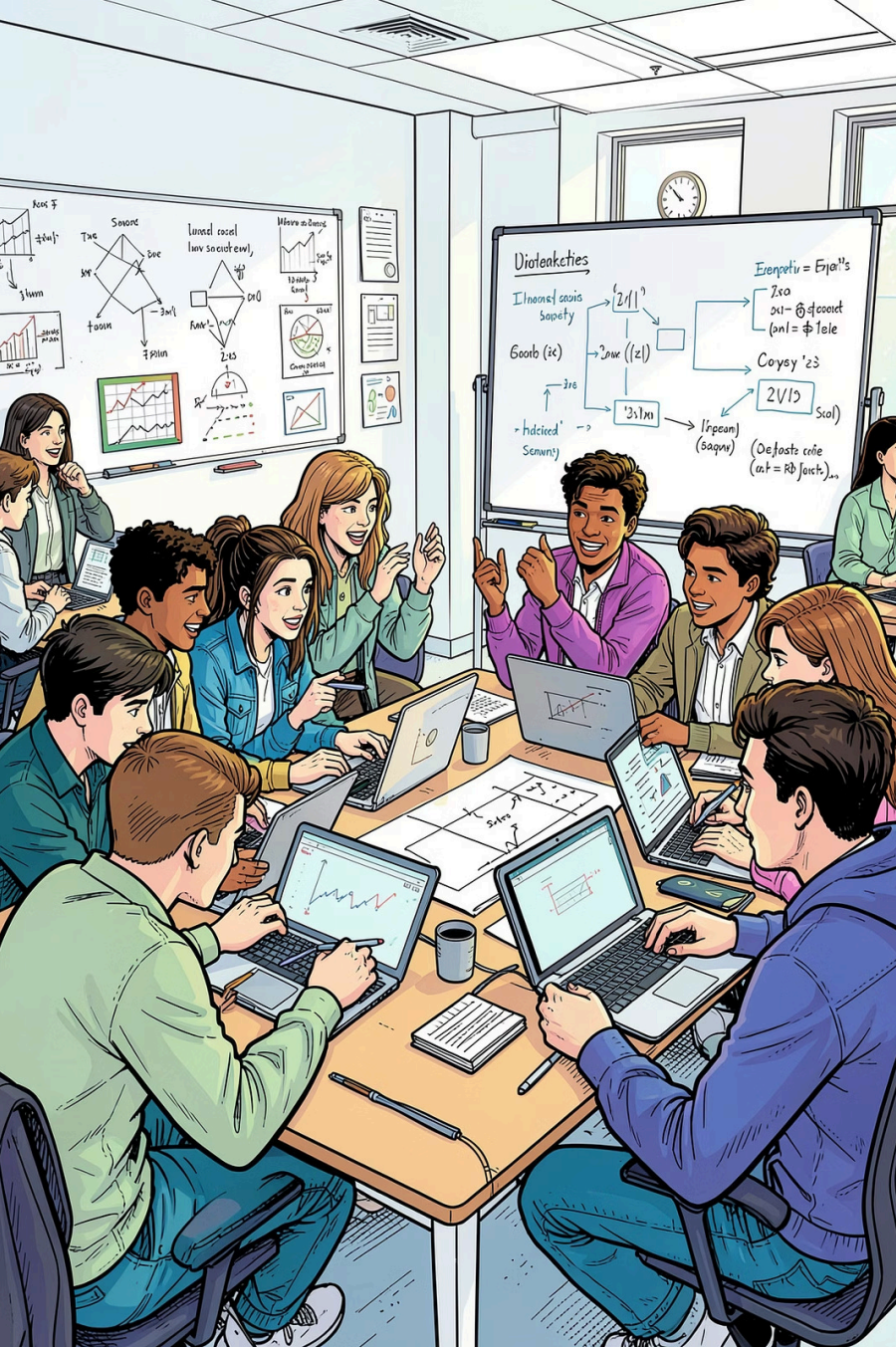
Road changes → sensors detect → AI decides → cycle continues

KEY DISTINCTION

ChatGPT vs. Agentic AI

An agent **augments** a language model with planning, memory, and tools – transforming it from a responder into an autonomous actor.

Capability	ChatGPT (LLM)	Agentic AI
Primary Function	Generates text responses	Completes multi-step tasks
Planning	None – reactive only	Strategic, multi-step planning
Memory	Limited to session context	Persistent, long-term memory
Tool Use	Very limited	Full API and tool integration
Workflow	Single prompt → single response	Multi-step autonomous execution



ACTIVITY & SUMMARY

Build a Hospital AI Robot

Activity: Design an AI robot for a hospital. Identify its **goal**, **environment**, **sensors**, **decisions**, and **actions**.

Intelligent Agent

Autonomous, goal-driven system

Sense → Decide → Act

The continuous agent loop

ChatGPT ≠ Agent

LLMs need planning, memory & tools

"AI becomes truly useful when it not only **understands** the world but also **interacts** with it intelligently – perceive, reason, decide, and learn from feedback."